



A Blockchain, Large Language Model, and Federated Learning based Secure Approach to Mitigate Grey Hole Attack in Software-Defined Vehicular Networks

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Abstract

Software defined vehicular networks (SDVN) allow vehicles to communicate effectively although they are susceptible to gray hole attacks, which are malicious nodes that use selective packet dropping to disrupt on-traffic and safety. The Traditional approaches that are in use are not completely effective or work well since they are not trustful, flexible, and offer real time protection in dynamic networks (dynamic network).

This project proposes a secure mechanism to locate and prevent gray hole attacks in SDVNs. The blockchain maintains an immutable, secure register of the actions of the nodes, thus everything is transparent and just. Large Language Models (LLMs) observe the patterns in the network in real-time to identify threats and generate intelligent solutions, such as new routing policies. Federated Learning allows vehicles to learn models collaboratively without exchanging personal information, enabling detection to be improved and privacy maintained.

There are three attack scenarios in the system. They are fixed rate selective dropping, time based intermittent dropping and target specific intelligent gray hole attacks. NS3 and SUMO will be used to perform simulations and Hyperledger Fabric will be used to implement the blockchain.

In this project, We will verify such key results as the ratio of packet delivery, latency and the reliability of the network in this project. This project aims at a helpful, secure and expandable system that will ensure SDVN communication is more dependable and secure to smart cities and self driving vehicles without malicious communication.

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Acronyms

AI	Artificial Intelligence
ANN	Artificial Neural Network
DAE	Deep Auto-Encoder
DSRC	Dedicated Short-Range Communications
E2E	End-to-End
FL	Federated Learning
GA	Genetic Algorithm
IDS	Intrusion Detection System
ITS	Intelligent Transportation Systems
KNN	K-Nearest Neighbor
LLM	Large Language Model
LSTM	Long Short-Term Memory
ML	Machine Learning
OSR	Open-Set Recognition
PDR	Packet Delivery Ratio
RF	Random Forest
RSU	Roadside Unit
SDN	Software-Defined Networking
SDVN	Software-Defined Vehicular Network
SVM	Support Vector Machine
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle
VANET	Vehicular Ad Hoc Network
VLC	Visible Light Communication

Chapter 1

Introduction

1.1 Background

1.1.1 Evolution of Vehicular Networks

Intelligent Transportation Systems (ITS) have brought a new generation of connected vehicles, which is able to communicate with each other and roadside infrastructure. Vehicular Ad Hoc Networks (VANETs) were the foundation of early vehicular communication systems and provided simple vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication. Nevertheless, VANETs have a number of challenges because of high vehicle mobility, high topology changes, and lack of centralized network control.

Software-Defined Vehicular Networks (SDVNs) have been proposed in order to address these shortcomings by extending the concept of software defined networking to motor vehicle contexts. SDVNs decouple the control plane and data plane to enable centralized controllers to make routing decisions, traffic flow, and resource allocation more efficiently. This is a programmable and flexible architecture that is needed to support safety-critical ITS applications like collision avoidance systems, traffic management, and emergency response services. [1, 2].

1.1.2 Security Challenges and Grey Hole Attacks in SDVN

Since SDVNs are becoming more and more significant in the context of road safety and autonomous driving, their safety and reliability have become one of the most significant concerns. The mobile nature, high velocity of vehicles and intermittent wireless connections are features of a dynamic topology that expose the vulnerability to security that cannot be completely addressed by conventional security measures.

The Grey Hole attack is one of the security threats that are hard to detect. In this attack, malicious nodes are used to selectively drop data packets as they pass through other nodes to ensure that the network looks normal. Such selective packet dropping may interfere with vital communication, including emergency alerts and collision warnings, without necessarily arousing suspicion. Consequently, the attacks of Grey Hole may cause a considerable decrease in network

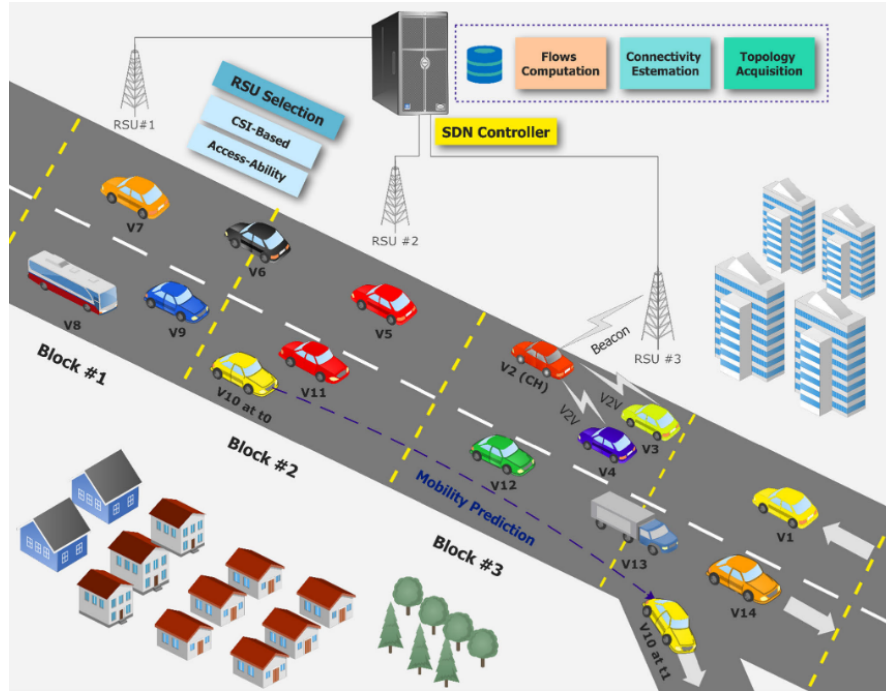


Figure 1.1: The basic principles of data routing in SDN-based VANET

performance and reliability that can be a serious threat to the safety of traffic and efficient transportation. [3].

1.2 Introduction to the Problem

One of the significant threats of SDVNs is grey hole attacks. In this attack, a set of data packets are selectively dropped by malicious vehicles as they forward them, and they look normal. This may interfere with crucial messages, including emergency messages, collision messages, and traffic messages. SDVNs are extremely mobile and have dynamic topology and it is quite impossible to detect such attacks. This issue is critical to the safe, reliable, and efficient communication on vehicles.

1.3 Limitations of Existing Work

The previous methods of ensuring vehicular networks are usually based on hard and fast rules, signature based detection, or simple machine learning. The traditional approaches find it difficult to identify selectively dropping attacks by the Grey Hole since the sporadic nature is similar to the normal network variations, and they are not able to respond well to the evolving traffic conditions. Moreover, most of the solutions need centralized data gathering or intensive computing, which pose privacy threats and scalability issues in dynamic vehicular settings [4, 5].

1.4 Problem Statement

Software-Defined Vehicular Networks can be attacked by selective dropping of Grey Hole attacks, in which malicious vehicles send packets at random to avoid being noticed. This behavior is very similar to the normal packet loss that occurs due to mobility and changing network conditions in high dynamic vehicular environments. Consequently, the current security strategies do not have effective and privacy-sensitive schemes to reliably differentiate between the Grey Hole attacks and the normal network operation, which restricts their accuracy and scalability in the real-world implementation of SDVN.

1.5 Objectives and Scope

1.5.1 Objectives

The objectives of this project are,

- In order to establish a comprehensive security system to identify and prevent opportunities to address both data plane and control plane Grey Hole attacks within Software Defined Vehicular Networks through Blockchain, Federated Learning and secure logging.
- To create privacy protecting detection schemes that integrate Blockchain, Large Language Models and Federated Learning, enabling vehicles to cooperate without transferring sensitive information.
- To check the efficiency of the suggested framework under realistic network conditions and prove its effectiveness over the current methods.

1.5.2 Scope

The project will utilize the simulation of a Software-Defined Vehicular Network (SDVN) and study of the attacks of the data plane and control plane, namely the Grey Hole attacks. It entails the use of Blockchain in documentation of vehicle interactions in a secure way and Federated Learning in training detecting models collaboratively and keeping vehicle information confidential. The performance of the framework in the aspects of detecting the attacks, network reliability, and efficiency is also analyzed as a part of the project. The evaluation of performance through simulation will be done on the basis of conventional network and security indicators, but actual implementation of the project in the real world and hardware installation is beyond the scope of the project.

1.6 Novelty

The originality of the research is that it has developed a single, privacy sensitive detection method against the attacks of the Grey Hole in Software-Defined Vehicular Networks. In contrast to the current solutions, the proposed framework

models packet-forwarding trust based on blockchain verified behavior, learns pattern of attack collaboratively under federated detection in the context of vehicular mobility, and approaches the problem of detecting attack as an adaptive learning problem based on Large Language Models, instead of rule-based mechanisms. Besides that, the work presents formal modeling of various behaviors of the Grey Hole attacks in realistic SDVN conditions, which allow proper detection in highly dynamic environments.

1.7 Contribution

This project makes a contribution to the field of vehicular network security through supporting scalable, privacy-sensitive, and methodologically novel infrastructure of detecting and addressing Grey Hole attacks in SDVNs. The main contributions are,

- **Blockchain based verification modeling:** Constructs a model in which blockchain is not only used to log data, but the mathematical verification of the packet forwarding behavior of the vehicles is performed for detecting suspicious nodes even in the case that attacks resemble normal traffic.
- **Federated Detection:** Federated learning is used to enable vehicles to jointly learn detection models, taking mobility, resource constraints, and local traffic dynamics into account, without sacrificing privacy and enhancing detectors in evolving settings.
- **Adaptive threat analysis using LLMs:** Invents the application of Large Language Models to scanned blockchain verified logs and identifies emerging trends in the pattern of the Grey Hole attacks, and does not perform attack detection based on pre determined rules.
- **Thorough attack modeling:** Formal models of various types of attacks to the Grey Hole have been studied: fixed-rate, time-based, and target specific under realistic vehicular network settings, which enables the targeted detection and mitigation of attacks.
- **Performance evaluation framework:** Conducts experiments to check the proposed framework in realistic SDVN settings and demonstrates good detection, minimal overheads, and resilience to various attacks of the SDVN under different behaviors of Grey Hole attackers.

1.8 Implication

This Project Suggests Implications to Vehicular Network Security, Studies, and Real Implementation.

- **Technological Innovation and Cross Domain Application:** Blockchain with Federated Learning is a novel method of making vehicle networks secure. The strategy can also be implemented in smart cities, traffic and other networks that are interconnected
- **Educational and Research Impact:** The project is a valuable demonstration of the contemporary security methods, as it enables the students and researchers to grasp the way to protect distributed networks.
- **Privacy Preserving Collaborative Security:** Federated Learning will allow vehicles to collaborate to determine attacks without any personal data; hence, the system is secure and scalable.
- **Improved Safety and Reliability:** The framework assists in ensuring good communication among the vehicles and minimizing the chances of vehicles causing accidents as well as aiding in managing traffic.
- **Foundation for Future Systems:** The project gives a road map on how to have secure, flexible and privacy-conscious vehicle networks that can be further applied to future transportation.

Chapter 2

Literature Review

Being centered around the convergence of blockchain technology, large language models (LLMs), and federated learning (FL), this literature review focuses on existing literature related to countermeasures of grey hole attacks in Software-Defined Vehicular Networks (SDVN). Owing to its dynamic network structure, a mobile network structure, and a distributed character of vehicular networks, routing integrity in SDVNs is vulnerable to grey hole attacks, which represent a sort of black hole attack as nodes might reject packets of interest selectively. SDVNs remain vulnerable to attacks on packet routing and availability since they adopt software-defined networks (SDN) and aim at building a centralized control structure alongside a distributed routing structure. The literature review is divided into sub-headings in relation to key tech: FL for securing and supporting a collaborative approach of handling large datasets, LLMs for superior threat analysis capabilities, as well as blockchain technology for a secure distributed network structure.

2.1 Blockchain in Vehicular Networks

Due to its unique decentralized and tamper-resistant properties, blockchain technology nowadays has been one of the great spotlights for enhancing the security issues in both VANET and SDVN. By storing immutable ledgers, it makes blockchain enable data integrity, transparency, and trust between involved vehicular nodes. In, a blockchain-based smart contract framework was presented in order to avoid black hole and grey hole routing attacks in VANETs. For the detection and isolation of malicious vehicles, this scheme utilized self-classification blockchain-based contracts (SCBC) and voting-classification blockchain-based contracts (VCBC). The network performance is assessed with packet delivery ratio (PDR) and throughput (TP) and the results showed good resiliency even in active attack conditions. It could be verified on node behavior, thanks to blockchain consensus mechanisms, without any trust in centralized authority [6].

In a different study, blockchain technology was integrated with federated Q-learning to provide a secure method of sharing data in VANETs. This system used extended elliptic cryptography (EX-ECC) for encryption and the interplanetary file system (IPFS) to provide decentralized file storage, with delegated practical Byzantine fault tolerance (DPBFT) used as the consensus algorithm. In order to provide

privacy in VANETs, decentralization of updating was used in a manner that limits its privacy in centralized architectures. The system showed high throughput and low latency in simulations conducted on up to 100 vehicles. In addition, the blockchain’s immutable ledger of transactions also provides support to ensure the tracing of evil acts, such as selective packet dropping in grey hole attacks, performed in VANETs. The major shortcoming of most studies, including VANETs, is that blockchain technology limits vulnerabilities associated with single point failures in software-defined VANET architectures, including those associated with software-defined network (SDN) controllers.

2.2 Decentralized Security through Blockchain and Federated Learning

A recent trend in SV research includes decentralizing the architecture of intrusion detection systems (IDS), which is prone to the problem of single points of failure and also a privacy concern as it involves the collection of data in a centralized manner. To counter these drawbacks, federated learning (FL) has gained popularity as a decentralized learning technique in VANETs. FL helps vehicles learn together to develop an IDS model from their respective generated data without revealing their vehicle details to others. To further protect against inference attacks, various studies have utilized differential privacy tools that involve the injection of controlled Gaussian noise [7].

At the same time, there has been an integration in FL frameworks utilizing blockchain technology, and more specifically, the use of consortium blockchains, to ensure trust and accountability among FL nodes. Specifically, consortium blockchains are employed to record and aggregate model updates into an immutable, transparent ledger, thus ensuring data integrity and unaltered data during their storage. Still, FL frameworks that support decentralized learning have a major deficiency, namely, they are susceptible to poisoning attacks carried out by malicious nodes that may provide compromised or deceptive updates. Recently, new frameworks, referred to as Zero-X, have been introduced, which include enhanced consensus protocols, such as Proof-of-Accuracy (PoA), to assess FL model updates based on their efficacy in enhancing global accuracy, thus making FL IDSs more resilient regarding decentralized learning configurations within highly dynamic vehicular environments.

2.2.1 Large Language Models in Security

Recently, research has emerged to apply the concept of Large Language Models (LLMs) to the security domain using models such as BERT (Bidirectional Encoder Representations from Transformers). In particular, attention has grown towards using models such as BERT in intrusion detection systems (IDS) with a focus on sequence-based anomaly detection in SDNVs. A research contribution specifically used a privacy-preserving framework for an IDS system utilizing Federated learning with BERT (FL-BERT) in software-defined VANETs. In this proposal for improvement, attack behaviors in terms of different attack patterns such as grey

hole attacks with patterns of intermittent packet drops have been incorporated by modeling network traffic logs as a sequence to capture dependencies in such patterns. Based on context provided by VeReMi, FL-BERT provided better accuracy for IDS over traditional ML methods such as Random Forest and Support Vector Machines by maintaining data locality via federated learning [8].

LLMs have been very effective in understanding semantic relations in network flow data, which helps in recognizing subtle routing anomalies in network flows to detect potentially selectively malicious activities as opposed to other kinds of threats. In correlated research studies, black hole attack prevention in VANETs has been done using threshold-based packet validation; nevertheless, this modeling has been found to have potentially vast improvement using LLMs with smart understanding capabilities for forged route request messages as well as anomalous control messages. In addition to this, edge models for unstructured or complex data representations like encrypted flow data patterns and obfuscated attack signs have been found to have immense applicability for LLMs as opposed to other models for determining corresponding data patterns in SDNs. Despite such benefits, research applications of LLM models in vehicular network security are still in their infancy and mostly involve offline process analysis based on their high computation as well as memory complexities. While potentially allowing real-time validation of flow rule data in SDNs by their collaboration with SDN controller models for grey hole attack validations, most correlated research models have failed to address edge device complexities in SDVN.

2.2.2 Federated Learning in VANETs

Federated learning (FL) has become a promising technique for jointly learning models in Vehicular Ad Hoc Networks (VANETs) and Software-Defined Vehicular Networks (SDVNs) to perform learning on models in a distributed manner without sharing data directly. A learning technique that is distributed in nature helps overcome the problem of intrusion detection systems related to user privacy in a vehicular network where accurate tracking is required for vehicles with proper location details without violating user privacy rules. A federated learning framework based on blockchain for VANET was designed using a blockchain-based federated Q-learning system for learning models based on Q-learning approach using deep Q-network learning models based on local observations and using InterPlanetary File System (IPFS) for model update off-chain storage with negligible time delay for threat detection with a high level of network throughput with an optimized blockchain architecture. This technique becomes significant for preventing Grey Hole attacks on a vehicular network where learning is performed using models based on federated learning with varying patterns during attacks based on different models for different vehicles based on different routes with accurate location tracking for preventing attacks on user privacy in a vehicular network [9, 10].

Another research effort introduced an efficient dynamic method for detecting black hole attacks in VANETs by analyzing forged route request packets and applying threshold-based calculations within the AODV routing protocol. This technique aligns well with the principles of federated learning, as FL supports continuous and distributed model updates across network nodes in response to changing at-

tack behaviors. In the context of software-defined VANETs, FL-BERT has been adopted to federate BERT-based intrusion detection models, enabling privacy-preserving training on misbehavior datasets such as VeReMi. This federated approach achieved higher attack classification accuracy compared to non-federated learning methods, demonstrating FL’s effectiveness in SDN-based vehicular environments. While FL significantly reduces the risks associated with data centralization in SDN architectures by keeping datasets local to each vehicle, it also introduces challenges, particularly communication overhead in high-mobility scenarios. Although existing studies primarily evaluate FL-based solutions in simulated VANET settings and report improvements in packet delivery ratio (PDR) and end-to-end delay, the combined integration of blockchain for tamper-proof model updates and large language models for advanced feature extraction remains largely unexplored, especially for SDVN-specific grey hole attack mitigation.

2.3 Gaps in Literature

- **Limited Technology Integration:** While most studies concentrate on individual technologies (e.g., blockchain for consensus or FL for privacy), very few integrate blockchain, LLMs, and FL into a cohesive framework for mitigating grey hole attacks in SDVN. This results in fragmented solutions that ignore synergistic advantages like secure, intelligent, and privacy-preserving detection.
- **Inadequate Attention to the Grey Hole Specificity:** While black hole attacks are frequently addressed, grey hole variants—which have selective and erratic packet dropping—are not as well studied. This is especially true in SDVN, where SDN controllers may be used to cause partial disruptions, leaving gaps in adaptive detection mechanisms.
- **Resource Constraints in Edge Devices:** Real-time applicability in dynamic SDVN environments is hampered by existing works that seldom take into account the computational overhead of LLMs like BERT in resource-limited vehicular nodes, frequently assuming unlimited processing power.
- **Absence of Real-World Datasets and Simulations:** The majority of evaluations rely on simulated datasets (such as VeReMi), with little testing conducted on actual SDVN deployments. As a result, factors that impact the effectiveness of attack detection, such as heterogeneous vehicle hardware, signal interference, and varying mobility patterns, are ignored.
- **Privacy vs. Performance Trade-offs:** FL-based methods protect privacy but frequently sacrifice model accuracy because of non-IID data distributions in VANETs; there are gaps in optimizing these trade-offs, particularly when combined with the overhead of blockchain technology and the complexity of LLMs.
- **Scalability in High-Mobility Scenarios:** Research demonstrates strong performance in small-scale simulations (e.g., 100 vehicles), but scalability to large SDVN with frequent topology changes is still unresolved, which could result in higher latency and consensus failures in blockchain-FL hybrids.

- **Lack of Holistic Threat Models:** The robustness of suggested solutions against combined or evolving threats is limited by the literature’s lack of comprehensive threat models that incorporate SDN-specific vulnerabilities (such as controller hijacking) alongside grey hole attacks.

Related Works Comparison

Table 2.1: Related Work Comparison Table

Ref.	Technologies Used	AI Models / Techniques	Methodology	Key Results	Limitations
Privacy-Preserving Intrusion Detection in Software-defined VANET using Federated Learning with BERT	Federated Learning, BERT, SDN, VANET	BERT for sequence classification, Random Forest, SVM, Logistic Regression, KNN	FL-BERT: Local training on VeReMi dataset, model aggregation for privacy-preserving IDS; compared with traditional ML models	FL-BERT outperforms traditional models in accuracy (up to 98% in related contexts), precision, recall, F1-score; enhances privacy and attack detection	Computationally intensive for BERT; evaluated in simulation only; potential for differential privacy in future work
An Efficient Dynamic Solution for the Detection and Prevention of Black Hole Attack in VANETs	AODV routing, NS-2 Simulator, VANET	Threshold-based detection (non-AI)	DPBHA: Calculates dynamic threshold from RREP sequence numbers, generates forged RREQ to detect and prevent BHA during route discovery	PDR increased by 3%, throughput by 6.15%, routing overhead reduced by 3.69%, E2E delay by 6.13%, detection rate 94.66%	Addresses only black hole attacks, not cooperative BHA or GHA; future work to include GHA and other DoS attacks
Mitigating Blackhole and Greyhole Routing Attacks in Vehicular Ad Hoc Networks Using Blockchain Based Smart Contracts	Blockchain, Smart Contracts, Ethereum (Remix IDE), VANET	Self-Classification (SCBC) and Voting-Classification (VCBC) Blockchain-Based Contracts	Smart contract-based routing: Replaces AODV control messages with SCBC/VCBC for node status classification to avoid malicious relays in highway scenarios	VCBC achieves PDR 78%, TP 1.16 kbps, RO 2.14%; outperforms SCBC (PDR 56%) and BCR (PDR 35%) even with black/gray hole attacks	High volatility in SCBC results; assumes highway stability; future enhancements for broader scenarios
A secure and efficient blockchain enabled federated Q-learning model for vehicular Ad-hoc networks	Blockchain, EX-ECC Encryption, IPFS, DPBFT Consensus, Federated Learning, VANET	Federated Learning	Data encryption with EX-ECC, local FL training, model aggregation; blockchain storage via IPFS, validation with DPBFT	Throughput 102465.8 KB/s, communication overhead 360.57 Mb, average latency 864.425 ms, encryption time 0.98 ms, decryption 1.97 ms, consensus delay 50 ms, validation 1.68 ms; superior to FedAvg, FPPDL, DRAM-BFL	Requires more complex datasets and security improvements; future work on computing capacity and efficiency

Chapter 3

Methodology

3.1 Proposed System Architecture

The architecture is divided into four functional layers so as to provide a secure and scalable vehicular network [11].

Data Plane (Vehicular Layer)

The SDVN has vehicles that communicate with each other and serve as data forwarders and Federated Learning clients. Every vehicle will contribute to the global security model by training on local data and preserving raw data confidential.

Control Plane (SDN Layer)

The SDN controller controls the flow of the network and communicates with the Blockchain and LLM modules. It uses blockchain information to authenticate routing conditions and implements adaptive policies that the LLM produced to isolate malicious nodes.

Blockchain Plane (Trust Layer)

The plane is dedicated, which offers a decentralized security basis:

- **Blockchain Ledger:** Serves as a non-modifiable, non-repudiable log of node packet forwarding actions to hold the organization accountable and to avoid the problem of attackers falsifying their logs. The ledger is independent of the analysis logic by isolating this to its own plane.

Knowledge and Security Plane (Intelligence Layer)

The analytical intelligence of the framework is in this plane:

- **LLM Agent:** Intelligently detects threats by forecasting and preventing the occurrence of the Grey Hole attacks by analyzing logs and reasoning semantically the importance of packets in real time.
- **Federated Learning Aggregator:** Federated Learning Aggregator: Aggregates the local vehicle model improvements across the globe to update the LLM without sacrificing privacy of the data [12].

3.2 Threat Model and Assumptions

3.2.1 Attacker Capabilities

- **Fixed-Rate Selective Dropping:** To cause network throughput disruption, malicious nodes drop an agreed percentage (e.g., 30–50%) of packets.
- **Time-Based Intermittent Attack:** Attackers alternate between normal and malicious states to mimic natural network fluctuations.
- **Target-Specific Intelligent Attack:** Nodes discard important packets (e.g. emergency broadcasts) depending on content or source or destination analysis.
- **Node Reputation Tampering:** Opponents can also seek to forge forwarding logs in order to escape [13].

3.2.2 Threat Model Assumptions

These assumptions define the constraints placed upon the adversary to bound the scope of the security framework:

- **T1 (Honest Majority):** The assumption is that most of the nodes involved in the blockchain consensus are honest, and this eliminates 51 percent attacks on the chain.
- **T2 (Cryptographic Bound):** The opponent is also presumed to be computationally constrained and is unable to create fake digital signatures or compromise the encryption of the blockchain.
- **T3 (Independent Malice):** This is the assumption that attackers do not cooperate to coordinate their intermittent dropping patterns so as to exactly replicate the traffic of a single honest node.

System Assumptions

- **A1 (Immutable Ledger):** It is assumed that the Blockchain provides a tamper-proof record of node behaviors.
- **A2 (Privacy Preservation):** Federated Learning is presumed to be effective to pool intelligence around the globe without revealing any single vehicle data.
- **A3 (Inference Constraints):** The approach assumes that the LLM can perform online adjustments based on blockchain data within the resource limits (latency and compute) of the vehicular environment.

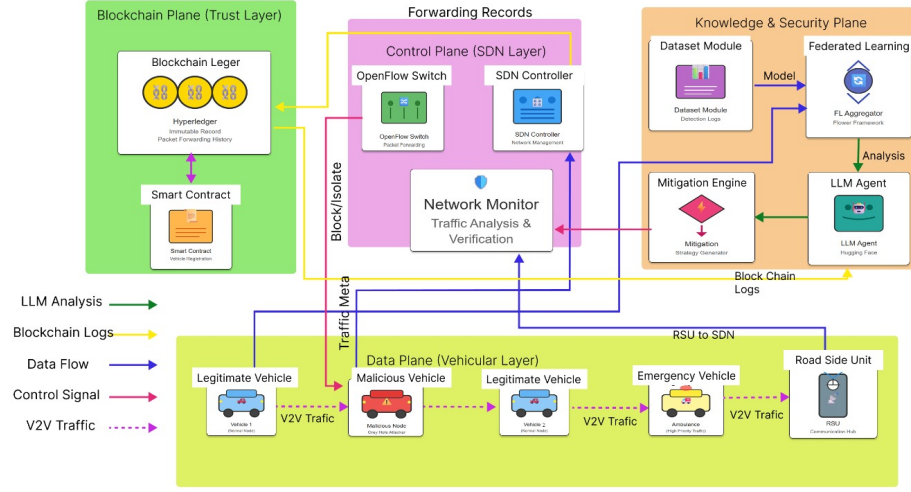


Figure 3.1: System architecture design for the proposed gray hole attack detection framework

Figure 3.1 shows the proposed solution outlines a secure Software Defined Vehicular Network (SDVN) solution that encompasses the use of blockchain, Large Language Models (LLMs), and federated learning for grey hole attack detection and mitigation. At the data plane, vehicles, both genuine, malicious, and emergency, exchange data via V2V and V2I communication, while malicious nodes randomly discard data in a manner that affects network performance. At the control plane, implemented by software-defined networking (SDN), OpenFlow switches, an SDN controller, and a network monitor oversee traffic flow centrally, detecting suspicious nodes in real-time. Records of forwarding data and traffic information are also stored in a secure manner in the blockchain plane, where an immutable ledger and smart contracts maintain data integrity, accountability, and genuine vehicle validation. Finally, the knowledge, or security, plane uses the blockchain transaction log for federated model training and includes the use of an LLM agent that interprets, in an intelligent, semantic fashion, the dynamic, and constantly evolving attack patterns, whose adaptive response strategies, by a mitigation engine, are implemented by SDN control signals, hence offering real-time, trustworthy, and intelligent grey hole attack protection.

The four layer architecture proposed works in unison to detect and isolate malicious nodes according to their distinct dropping patterns. The mitigation of the variants of the Grey Hole is made possible by the following components:

- **Data Plane Mitigation (Target-Specific Attacks):** The Vehicular Layer gathers real-time Traffic Meta data, such as packet headers and priority levels (e.g. differentiating an Emergency Vehicle and normal nodes). Such granular visibility enables the system to identify Target-Specific Intelligent attacks in which an attacker chooses to drop important safety messages and send normal data. Road Side Unit (RSU) is an intermediary, which makes sure that this telemetry is sent to the control plane to be analyzed immediately.
- **Control Plane Mitigation (Real-time Isolation):** The Network Monitor performs continuous traffic verification by comparing the Traffic Meta from the Data Plane against the Forwarding Records. When the SDN Controller identifies a discrepancy such as a node dropping 30%–50% of traffic (Fixed-Rate Dropping) it issues a Control Signal ("Block/Isolate") to the OpenFlow Switch. This enables the network to dynamically reroute traffic around the malicious node in real-time.
- **Blockchain Plane Mitigation (Reputation and Record Integrity):** The Blockchain Ledger (Hyperledger) keeps a record of packet-forwarding that cannot be altered to prevent Node Reputation Tampering. Since these Blockchain Logs cannot be modified by attackers, the system has a ground truth to detect Fixed-Rate Selective Dropping over long periods of time. The Smart Contract will guarantee that only nodes with a valid, unspoiled history will be trusted to use in the future, effectively negating any effort to spoof forwarding logs.
- **Knowledge & Security Plane Mitigation (Intermittent and Complex Patterns):** The LLM Agent conducts profound Analysis of blockchain records and network trends in order to detect Time-Based Intermittent attacks that switch between malicious and honest conduct to simulate network variations. An attacker may be able to circumvent a simple threshold-based detector, but the Federated Learning (FL) Aggregator utilizes the Flower Framework to integrate local detection models across a group of vehicles. Such cooperative intelligence enables the Mitigation Engine to develop dynamic strategies that anticipate and prevent advanced attackers prior to them causing havoc to the network.

The system is proposed with a unique high-level functional pipeline that is expected to convert raw network telemetry into actionable security intelligence of Software-Defined Vehicular Networks (SDVNs). This process begins with multi-dimensional feature extraction where raw packet logs, link status and vehicle positions are transformed to produce a topology graph. This allows Structural Consistency Analysis (GAT) to compute a structural suspicion score, which measures nodes with behavior that is not according to the physical and logical network patterns, and at the same time computes temporal and QoS metrics to perform deep contextual analysis.

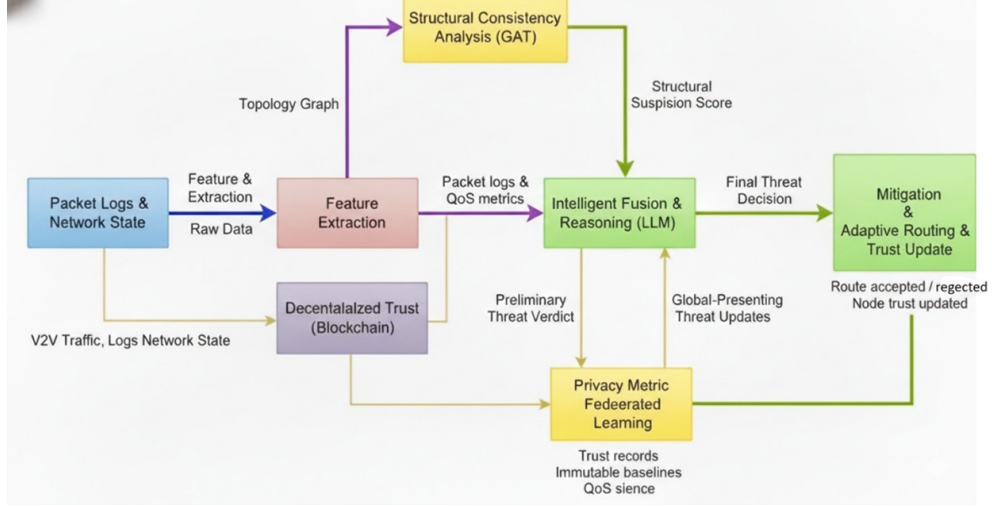


Figure 3.2: High-level block diagram of the proposed blockchain-, federated learning-, and LLM-enabled secure SDVN architecture

The Intelligent Fusion and Reasoning (LLM) module is at the center of the framework and is a centralized semantic decision engine. This block has a preliminary threat verdict that combines the data of two important verification channels: the Historical Metric Verification (Blockchain), which compares the present behavior and immutable baselines and trust records, and the Privacy Metric Federated Learning block that provides updates on threats to the whole world, without jeopardizing the privacy of local data. This structural, historical and collaborative intelligence will enable the system to detect with ease advanced variants of the Grey Hole which usually cannot be detected using the conventional detection techniques.

The last functional block is the Mitigation, Adaptive Routing and Trust Update block which implements final threat decision. When a malicious node is detected, the system initiates adaptive routing policies, blocking compromised paths and recalculating safe paths, so that the delivery of high-priority traffic, such as emergency broadcasts, can proceed. More importantly, the system completes the security loop by retransmitting a "Node Trust Updated" message to the decentralized trust layer, which guarantees that the global reputation of the node is updated in the unalterable blockchain ledger to ensure that the reputation of the node in the future is not compromised.

3.3 Attack Scenarios in Gray Hole attack

In this project, we consider three types of Grey Hole attacks in Software Defined Vehicular Networks under both controller plane and data plane attack scenarios.

3.3.1 Data Plane version attack scenarios

Fixed Rate Selective Dropping Grey Hole Attack in Data Plane

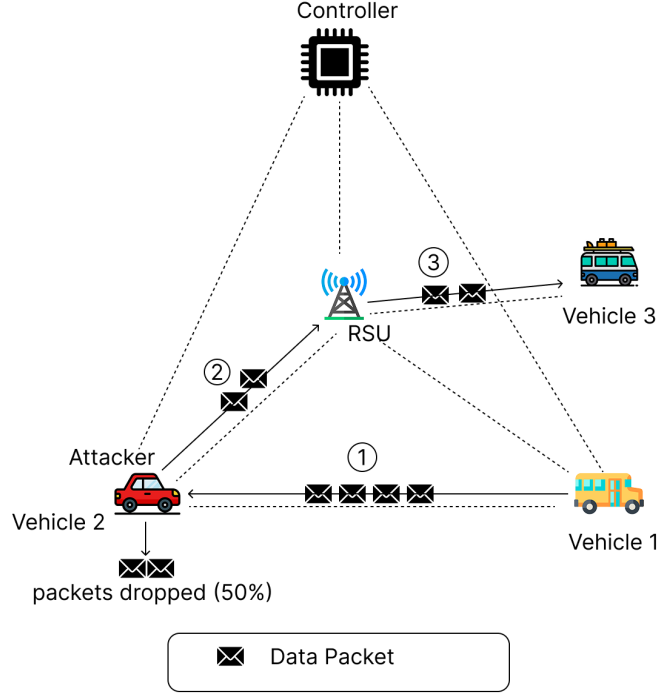


Figure 3.3: Fixed-rate selective dropping Grey Hole attack in the data plane

Figure 3.3 shows a fixed rate selectively dropping grey hole attack of the data plane. According to that figure there are four vehicles and one RSU in that network. In this network, vehicle 2 act as a attacker.

As shown in figure 3.3 the attacker begins packet dropping by 50% , data packet came from vehicle 1 to vehicle 2 (attacker) as step 1. Then after attacker (vehicle 2) forward remain 2 packets to the RSU as step 2. Finally, figure also show the two packets that came from attacker forwards to the vehicle 3.

Time Based Intermittent Grey Hole Attack in Data Plane

Figure 3.4 and 3.5 show the time based intermittent gery hole attack scenario in data plane two different times. In this figure there are three vehicles , one RSU and a SDN controller.

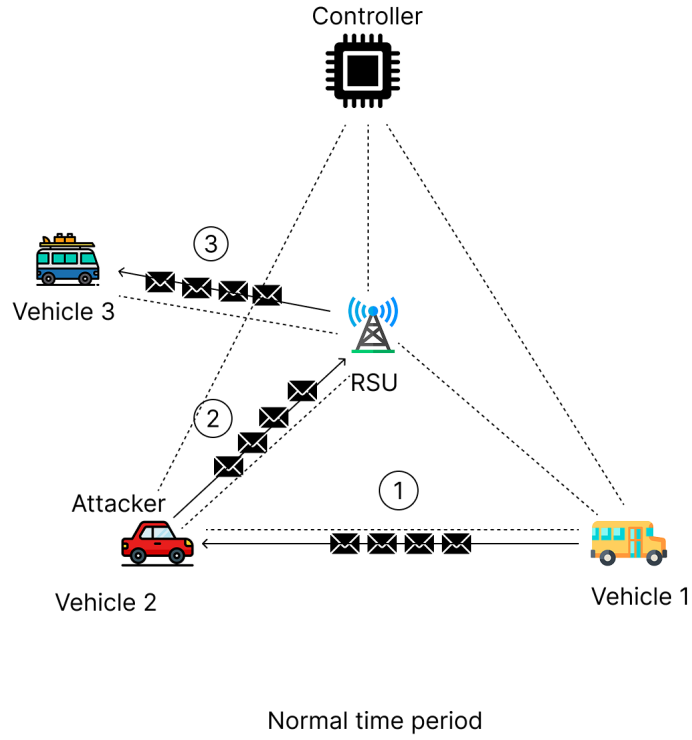


Figure 3.4: Normal data-plane behavior at time t_1 with no packet dropping

Figure 3.4 shows the normal behavior of the data plane at time $t = t_1$. According to the figure, all vehicle act as correctly without packet dropping. As a first step vehicle v1 sends 4 packets to the vehicle 2 (attacker) , but that time attacker act as normal vehicle without packet dropping. Then after vehicle 2 forward all four packets to RSU as a step 2. As a final step (step 3), RSU forward 4 packets to the vehicle 3. So, during this period communication proceeds normally without packets dropping.

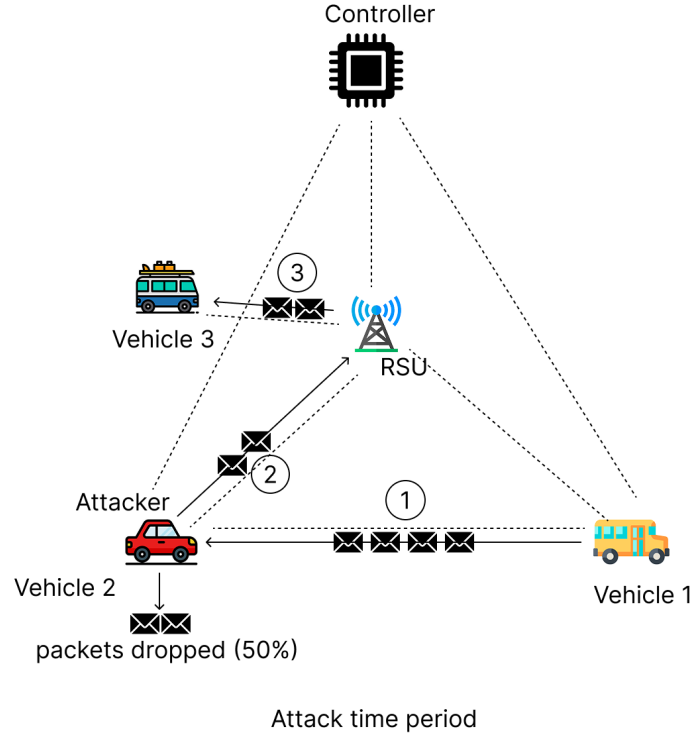


Figure 3.5: Malicious data-plane behavior at time t_2 with intermittent packet dropping by V2

As shown in figure 3.5, at time $t = t_2$, attacker act as a malicious with selectively packets dropping in the network. As a first step, vehicle sends 4 packets to the vehicle 2 (attacker). Then after attacker act as malicious node with selectively packets dropping. So, attacker drop two packets (50%) and forward remain two packets to the RSU as a second steps. Finally, RSU forward that two packets to the vehicle 3 as a third step. So, like these attack scenarios, it is difficult to detect the attack due to changing malicious behavior.

Target Specific Intelligent Grey Hole Attack in Data Plane

Figure 3.6 shows the target specific intelligent grey hole attack scenario in data plane. In this figure there four vehicles, one RSU and a controller. Also figure shows using dots line flow rule installed by the controller in that vehicle network. So, in that network vehicle 2 act as attacker.

As shown in Figure 3.6, vehicle 3 sends three packets to the vehicle 2 (attacker) as a 1st step and attacker forward all that three packets to the RSU without any dropping as 2nd step. in that path, step three show, that three packets forward to the vehicle 4. So that path work correctly according to the flow rule. 4th step in figure show another path that forward four packets to the vehicle 2 (attacker). So, attacker drop 50% packets and forward only remain two packets to the RSU as a 5th step. Finally, RSU forward that two packets to the vehicle 4. As conclusion of this figure attacker target only specific vehicle or vehicles to drop packets in a

vehicular network.

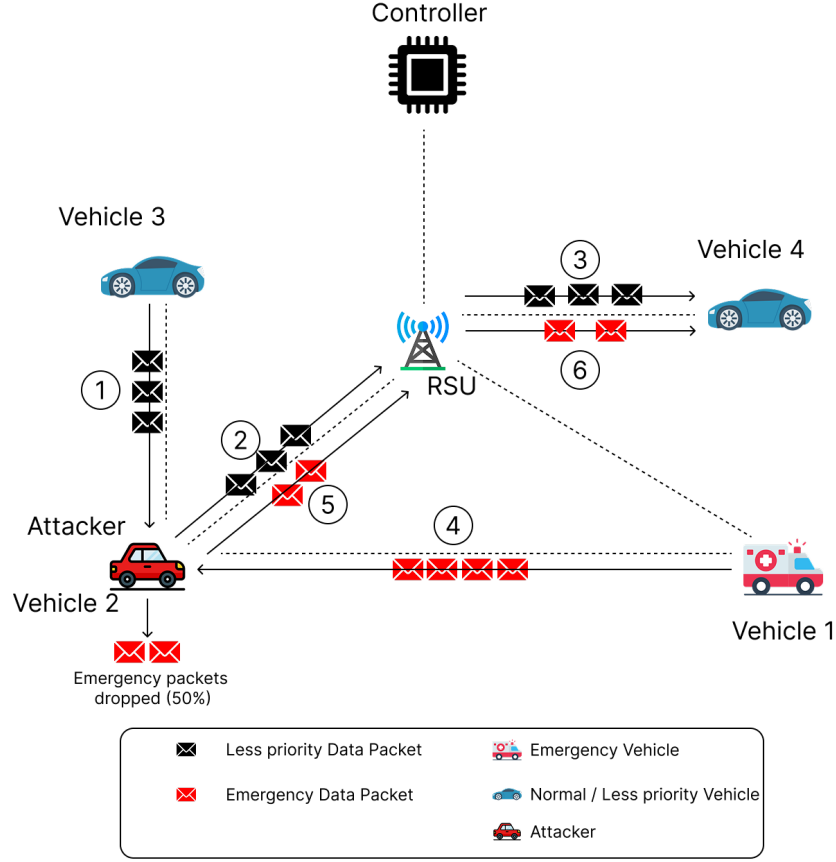


Figure 3.6: Target-specific intelligent Grey Hole attack in the data plane

3.3.2 Controller Plane version attack scenarios

Fixed Rate Selective Dropping Grey Hole Attack in Controller Plane

Figure 3.7 show the fixed rate selectively dropping grey hole attack scenario in the controller plane. So, in this controller install malicious flow mod rule in the vehicular network act as a attacker.

As shown in Figure 3.7, first controller install malicious flow mod rule on RSU and the vehicle 2. So, controller configure on the RSU to drop 50% of all incoming flows and configures on vehicle 2 to drop 60% of its received packets. 2nd step show vehicel 1 send 5 packets to the vehicel 2 and 3rd step show, vehicle 2 drop three packets according to the malicious configuration. In 4th step send two packets to the vehicle 3 and vehicle forward all two packets to the RSU as a 5th step. Since RSU already configured malicious flow mod rule RSU drop 50% packets and forward remain packet to the vehicle 4 as 6th step. Since packets are only partially dropped, the communication appears normal, while the overall network performance is gradually degraded also making the attack difficult to detect in like these dynamic vehicular network.

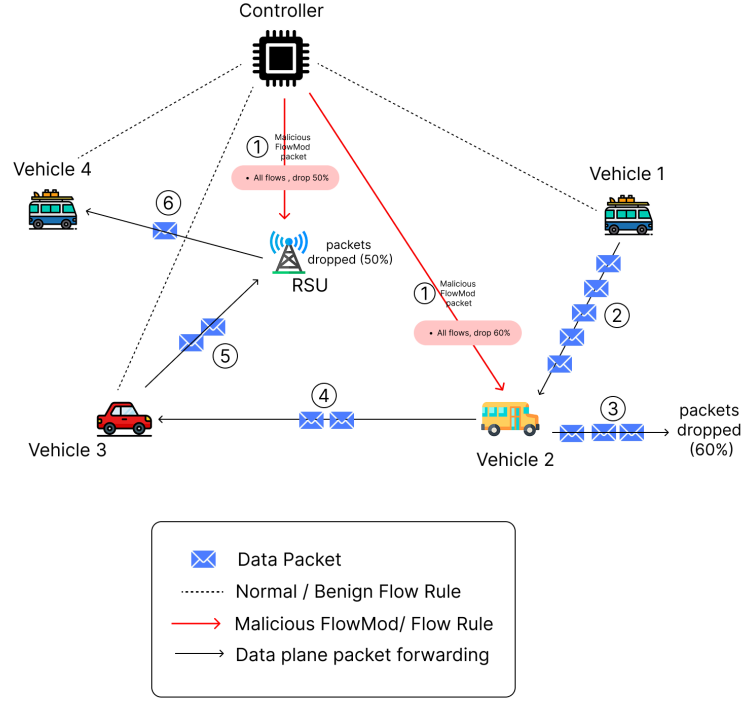


Figure 3.7: Controller plane fixed rate selective dropping Grey Hole attack

Time Based Intermittent Grey Hole Attack in Controller Plane

Figure 3.8 and 3.9 show the time based intermittent gery hole attack scenario in control plane two different times.

As shown 3.8 all steps working without packet dropping according to the flow rule. at time $t = t_1$, the SDN controller behaves normally and does not perform any packet dropping. The controller installs correct flow rules on all vehicles and the RSU.

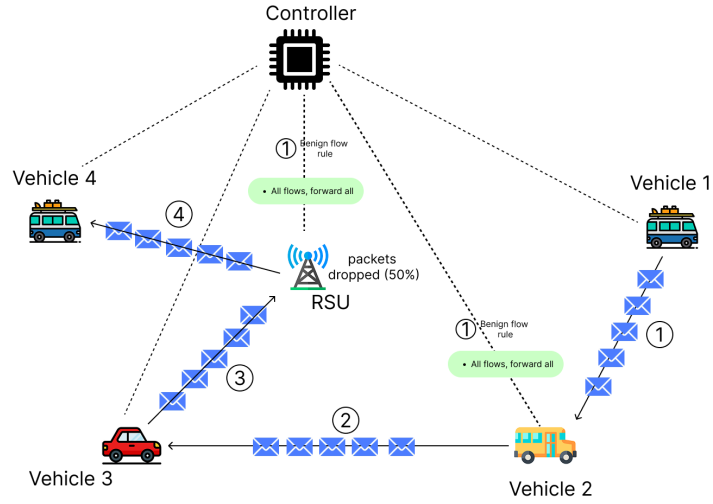


Figure 1 : $t = t_1$ time, Normal behavior of controller (Without packet dropping)

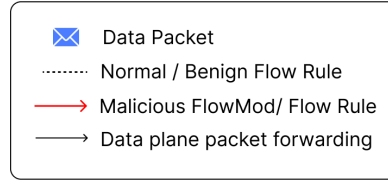


Figure 3.8: Normal controller behavior at time t_1 with no packet dropping.

As shown figure 3.9 at time $t = t_2$, SDN controller install malicious flow mod rule in the network. So, controller configure on the RSU to drop 50% all flows and on the vehicle 2 to drop 60% all flows. Then as the 2nd step vehicle 1 send 5 packets to the vehicle 2 and vehicle 2 drop 60% packets and forward remain two packets to the vehicle 3 as the 4th step. In 5th step show vehicle 3 forward that two packets to the RSU. According to the malicious flow mode rule on RSU, RSU drop one packet and forward remain packet to the vehicle 4 as the 6th step. This intermittent behavior allows the controller to hide the attack by switching between normal and malicious operation.

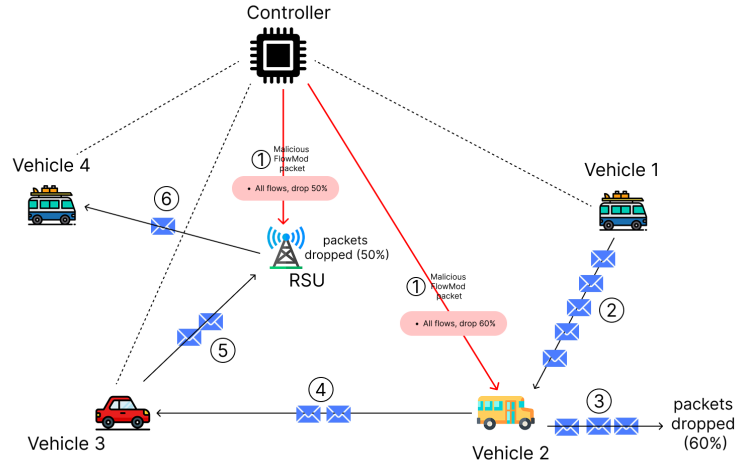


Figure 2 : $t=t_2$ time, Malicious flow rules are installed by controller

Figure 3.9: Malicious controller behavior at time t_2 with selective packet dropping.

Target Specific Intelligent Grey Hole Attack in Controller Plane

Figure 3.10 shows the target specific intelligent grey hole attack in controller plane. So, in this first controller install flow rule. So, in this figure only install malicious flow mod rule on RSU. So, RSU drop 50% packets only if source is vehicle 1. All packets came from other sources, forward without any dropping.

As shown 3.10, first step install malicious flow mod rule on RSU. 2nd step shows, vehicle 1 send 4 packets to the RSU and 3 rd step shows, RSU drop two packets since source is vehicle 1. After RSU forward remain 2 packets to the vehicle 4. In step 5 and 6 show the another path packet routing, in that steps there is no any packets dropping since in that path not contain source vehicle 1. In conclusion of this attack scenario, by selectively targeting emergency traffic while forwarding normal traffic, the controller degrades safety critical communication while remaining difficult to detect.

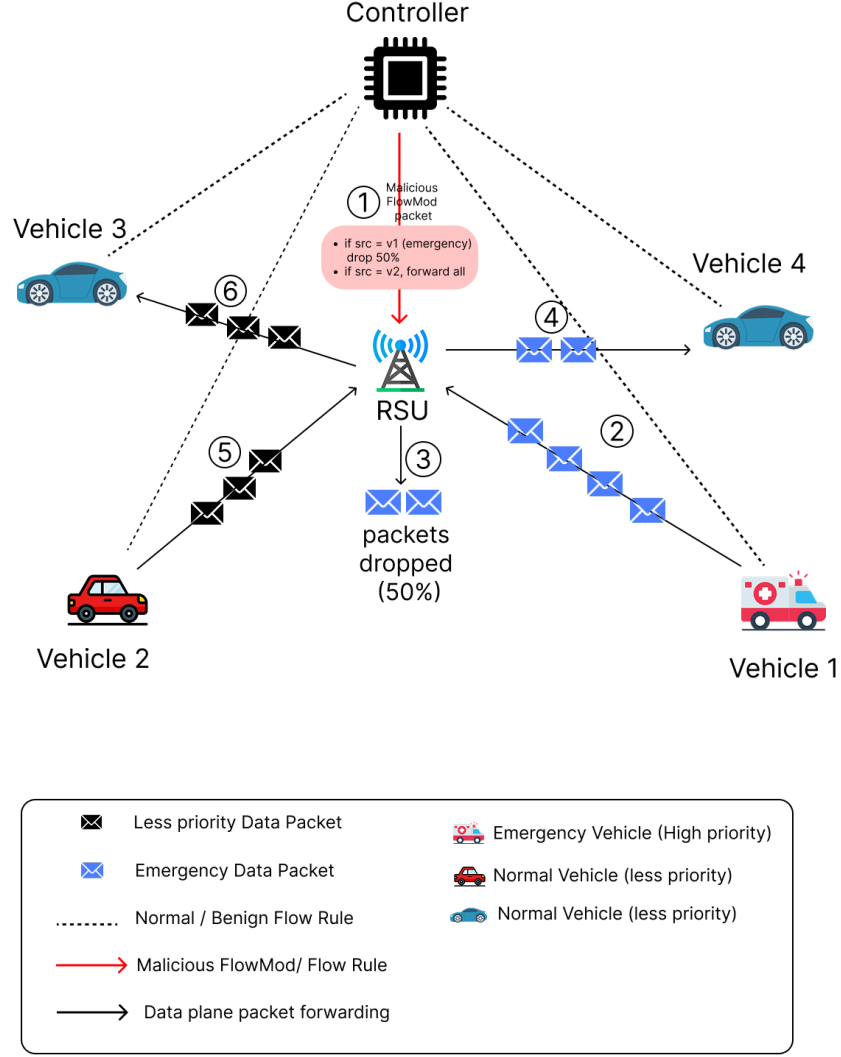


Figure 3.10: Target-specific intelligent Grey Hole attack in the controller plane

3.4 Research Design

The study adheres to experimental and simulation based research design to come up with decentralized security framework of Software-Defined Vehicular Networks (SDVNs). The strategic objectives are to reduce three different variants of attacks on the Grey Hole, namely Fixed-Rate Selective Dropping, Time-Based Intermittent, and Target-Specific Intelligent attacks.

The suggested solution presents a multi-layered defense system that involves Blockchain, Large Language Models (LLMs), and Federated Learning (FL). The methodology overcomes the shortcomings of conventional defenses in high mobility settings by offering real-time flexibility and decentralized trust by the following combined system elements [14],

- **SDVN Simulation and Attack Modeling:** An SDN-based architecture that can be used to handle high-mobility situations by adding selective packet-dropping detection features to detect various attack patterns in real time.
- **Decentralized Trust via Blockchain:** A distributed ledger, developed based on Hyperledger, that stores an immutable record of behaviors of nodes to provide transparency and stop reputation manipulation in packet forwarding operations.
- **Intelligent Threat Detection with LLMs:** A module based on LLM and complex network pattern analysis and blockchain logs to forecast Grey Hole signatures and produce adaptive routing policies by semantic reasoning.
- **Privacy-Preserving Federated Learning:** To enhance scalability and security, Federated Learning is implemented to train detection models collaboratively across vehicles without centralizing sensitive traffic data.

3.5 Data Collection and Simulation Environment

The limited real world datasets to use in vehicular Grey Hole attack makes this study use a controlled data generation process in a simulated environment. The area of data collection is centered on the dynamic interactions between the high-mobility vehicular nodes and the multi-layered security framework.

3.5.1 Data Generation Process

The simulation environment is set to simulate high-mobility vehicular situations and the standard communication protocols needed by Software-Defined Vehicular Networks. To have a strong dataset to train the detection, the environment adds malicious behaviors that are programmed to simulate three different variants of the attack of the Grey Hole:

- **Selective Dropping Data:** Traffic flows of different percentages of packet-loss are generated to model fixed-rate attacks.
- **Temporal Discrepancy Data:** Simulation of intermittent malicious states to generate datasets that are time-based attacks.
- **Semantic Traffic Data:** Giving priority to certain type of packets like emergency broadcasts to emulate target-specific intelligent attacks.

3.5.2 Functional Feature Extraction

Raw simulation logs are analyzed to capture certain features that are needed by the decentralized security modules:

- **Behavioral Logs:** Record documents of packet transmission and reception success that can be used to check integrity of nodes over time.

- **Network Contextual Features:** This is used to identify complex behavioral patterns like Temporal and semantic metadata, including packet timestamps and priority headers.
- **Stochastic Model Gradients:** Locally computed model updates for each vehicle's, and also extracted for collaborative learning without compromising raw data privacy.

3.6 Performance Evaluation

To prove the efficiency of the suggested multi-layered security framework, it is compared to three state-of-the-art (SOTA) techniques that are the representatives of various paradigms of the Grey Hole detection in the vehicular networks. The choice of these comparison methods is due to their recent publication and the relevance to individual elements of our framework (Neural Networks, Blockchain, and Machine Learning):

- **ANN-ABC based IDS (2023):** A smart Intrusion Detection System, based on an Artificial Neural Network (ANN) optimized using the Artificial Bee Colony (ABC) algorithm to detect packet-dropping patterns in AODV-based vehicular networks.
- **VCBC Smart Contract Protocol (2024):** A security mechanism that is based on the use of Voting-Classification Blockchain-Based Contracts (VCBC) to categorize the nodes as white, black, or grey according to their ratios of packet delivery that is documented on a public ledger (based on blockchain).
- **RF-SVM Anomaly Detection (2023):** An enhanced Machine Learning-based IDS utilizing a Random Forest (RF) classifier to identify subtle Grey Hole signatures through an analysis of 20 network traffic features derived out of simulation logs.

3.7 Performance Evaluation Metrics

The proposed and baseline techniques are tested with the help of a set of quantitative metrics that will measure the security performance and the operational performance on the network. The metrics undertaken are the following:

3.7.1 Network Performance Metrics

- **Packet Delivery Ratio (PDR):** The proportion of the data packets that have been effectively sent to the target destination nodes relative to the total amount of packets sent. This is the main measure of the capability of the framework to counter the attacks of packet-dropping.
- **Throughput:** The rate of successful data transmission over the network, measured in bits per second (bps) or bytes per second (Bps). This evaluates the overall network efficiency under attack conditions.

- **End-to-End (E2E) Delay:** The average time taken for a packet to travel from the source vehicle to the destination node. This measure evaluates the delay caused by the multi-layered analysis (Blockchain consensus and LLM inference).

3.7.2 Detection Reliability Metrics

- **Detection Accuracy:** The proportion of the number of correctly identified malicious and legitimate behaviors to the total number of test samples, which gives a general picture of the model performance.
- **False Positive Rate (FPR):** The percentage of legitimate nodes wrongly identified as malicious to the overall legitimate number of nodes. This is an important metric in SDVNs to make sure that security measures do not introduce an unwarranted fragmentation to the network.
- **Matthews Correlation Coefficient (MCC):** An effective statistical rate that yields a high score can only be achieved when the detection results are good in all the four categories of the confusion matrix (True Positives, True Negatives, False Positives, and False Negatives). This is used to assess the performance of the LLM and FL modules, namely in unbalanced datasets where malicious nodes are not common as the legitimate nodes.

Chapter 4

Timeline and Resource Required

4.1 Timeline

The timeline provides an overview of project tasks scheduled for December 2025 to August 2026, showing key project milestones along with their expected durations in weeks.

- **MILESTONE 01: Project Topic Selection**
Tasks: Select topic, read proposal description, get supervisor approval.
- **MILESTONE 02: Supervisor Discussion**
Tasks: Discuss project feasibility and approach.
- **MILESTONE 03: Literature Review**
Tasks: Study SDVN, Grey Hole attacks, blockchain, and related work.
- **MILESTONE 04: Research Gap Analysis**
Tasks: Identify gaps and limitations in existing work.
- **MILESTONE 05: Proposal Preparation**
Tasks: Draft and finalize project proposal.
- **MILESTONE 06: Technology Study**
Tasks: Study NS3, SUMO, Blockchain, ML tools.
- **MILESTONE 07: System Architecture Design**
Tasks: Design overall system structure.
- **MILESTONE 08: Threat Model Design**
Tasks: Identify possible Grey Hole attack scenarios.
- **MILESTONE 09: NS3-SUMO Setup**
Tasks: Install and configure simulation environment.
- **MILESTONE 10: Basic SDVN Simulation**
Tasks: Simulate SDVN without attacks.
- **MILESTONE 11: Blockchain Implementation**
Tasks: Deploy Hyperledger Fabric for node logging.

- **MILESTONE 12: LLM Integration**
Tasks: Integrate LLM for monitoring network logs.
- **MILESTONE 13: Federated Learning Setup**
Tasks: Configure FL across simulated nodes.
- **MILESTONE 14: Attack Scenarios Modeling and Simulation**
Tasks: Implement Grey Hole attacks and test mitigation.
- **MILESTONE 15: System Integration and Testing**
Tasks: Integrate all components and run tests.
- **MILESTONE 16: Performance Evaluation**
Tasks: Analyze detection accuracy, latency, and network performance.
- **MILESTONE 17: Final Documentation**
Tasks: Write final report and prepare diagrams.
- **MILESTONE 18: Final Presentation and Viva Preparation**
Tasks: Prepare slides and rehearse presentation.

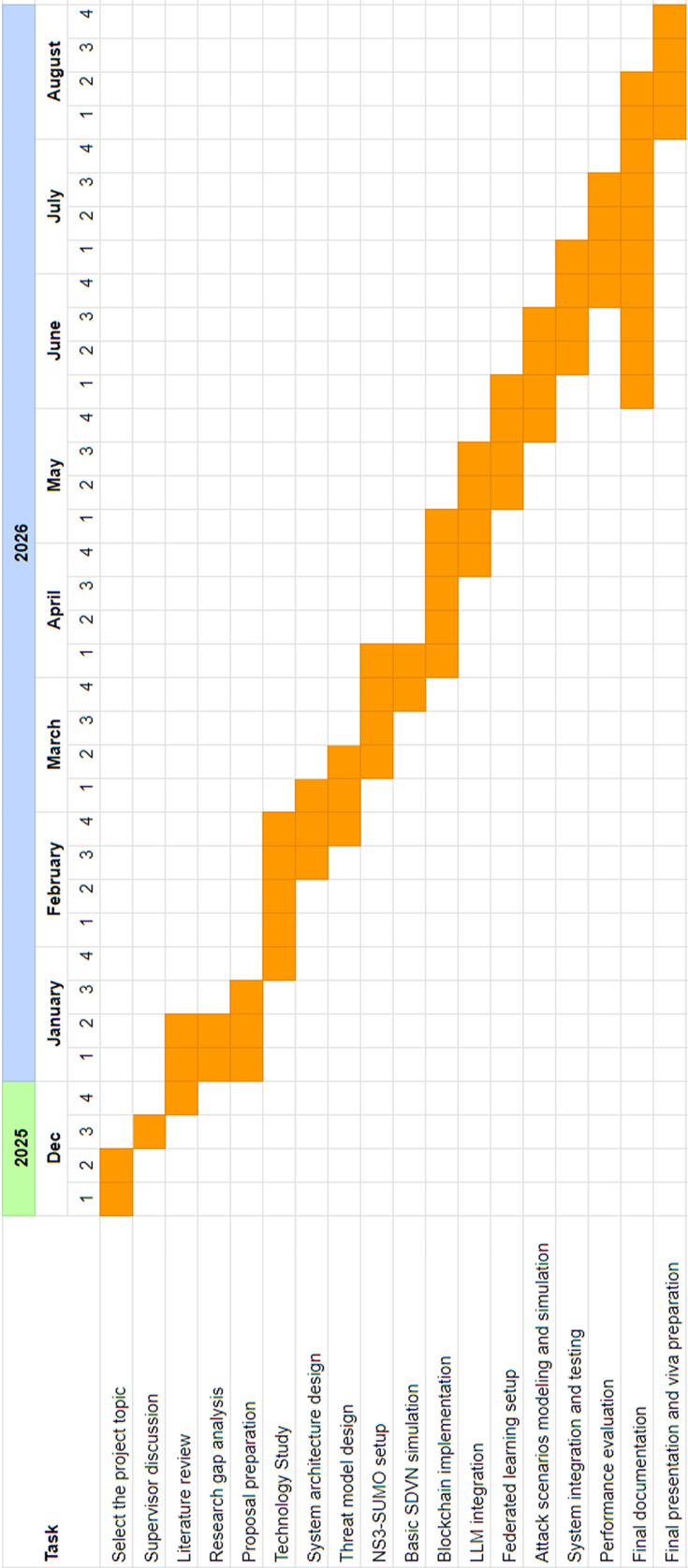


Figure 4.1: Project timeline from December 2025 to August 2026.

4.2 Resources Required

This section provides a succinct description of the hardware and software tools that are necessary to complete the proposed project successfully on detecting and mitigating gray hole attacks in software defined vehicular networks

4.2.1 Hardware Resources

- **Computer / Workstation:** A system with at least an intel i7 or Ryzen 7 processor and a minimum of 16GB RAM also sufficient storage to execute network simulations, blockchain components, and machine learning tasks.

4.2.2 Software Resources

Network Simulation Tools

- **NS-3 (Network Simulator 3):** Used to simulate the Software-Defined Vehicular Network environment and Grey Hole attack scenarios.
- **SUMO (Simulation of Urban Mobility):** Used to generate realistic vehicular mobility patterns for the simulations.
- **C++ and Python:** Used for NS-3 scripting, simulation control, and pre-processing simulation data.

Machine Learning and Security Tools

- **Python:** Used for implementing machine learning models and processing simulation data.
- **TensorFlow / PyTorch:** Used for developing and training learning models.
- **Hugging Face Transformers:** Used for integrating Large Language Models (LLMs) to analyze network behavior and assist in Grey Hole attack detection and response.
- **Flower Framework:** Used to implement Federated Learning for distributed, privacy-preserving model training across vehicular nodes.
- **PyCryptodome:** Used for cryptographic operations and security-related functions.

Blockchain Platform

- **Hyperledger Fabric:** Used to implement a secure and tamper-proof distributed ledger for logging node behaviors.
- **Docker:** Used for containerized deployment of blockchain network components.

Analysis and Visualization Tools

- **Python Libraries (NumPy, Pandas, Matplotlib):** Used to analyze simulation results and visualize performance metrics.
- **Wireshark (Optional):** Used for packet-level analysis and validation.

Version Control and Documentation

- **GitHub:** Used for version control and collaboration among team members.
- **Visual Studio Code / PyCharm:** Used as integrated development environments for coding and debugging.
- **LaTeX / Overleaf:** Used for preparing the project proposal, reports, and final documentation.
- **Draw.io / Lucidchart:** Used for designing system architecture and workflow diagrams.

Chapter 5

Conclusion

The project suggests an innovative and feasible solution to minimize the grey hole attacks in the software defined vehicular networks. Grey hole attacks are also hard to detect since malicious vehicles selectively drop packets in various ways, including dropping packets only at specific times or specific messages, and most of ways. Thus, such types of attacks can have a severe impact on road safety and network performance of vehicular network. This project also considers scenarios where the network controller may be malicious and place malicious flow rules between the vehicle network.

To address this issue in vehicular network, the project integrates three current technologies which are Blockchain, Large Language Models, and Federated Learning. The packet forwarding activities are safely recorded using blockchain to ensure that no behavior of the nodes is modified or concealed. Large Language Models are useful in network data analysis and detecting abnormal patterns that point to Grey Hole attacks. Federated Learning enables vehicles to jointly train detection models without exchanging their own data, and enhances privacy and scalability of a distributed network.

Simulations will be used to assess how well the system is able to detect various types of attack by Gray Hole and ensure that the network communication is not disrupted. The combination of security, learning, and analysis techniques is likely to enhance the trust between vehicles. The project will also assist in learning how the modern technologies can be used to address the real world security issues in the vehicular networks.

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